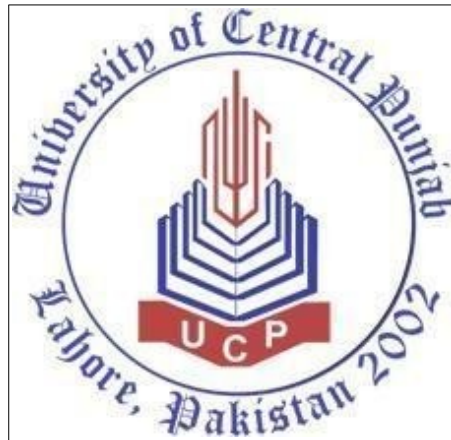


Introduction to Database

Semester Final Project

Pakistan Census Database

Session: Spring 2026



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1. Project Title

Pakistan Census Database

2. Project Overview

The **Pakistan Census Database** is a centralized system built to collect and manage national demographic, household, and geographic data.

- ◆ **Geographic Tracking:** Organizes data down a strict chain from Provinces and Districts to local Census Blocks.
- ◆ **Property & Housing Utilities:** Tracks the quality of physical structures and access to water, power, and fuel.
- ◆ **Population Registries:** Stores citizen profiles, including CNICs, family structures, literacy, and income.
- ◆ **Field Operations:** Manages the census workforce by linking local enumerators to their regional supervisors.

Ultimately, this system provides fast, reliable data slicing to help the government make accurate decisions regarding resource allocation and national planning.

3. Objectives

Objective 1: Securely store and quickly find data across all geographic levels, from whole provinces down to local census blocks.

Objective 2: Track citizen details accurately, including CNICs, family structures, income, literacy, and housing conditions.

Objective 3: Support generating clear population reports, literacy rate summaries, and regional development statistics for government planning.

Objective 4: Perform complex SQL queries to filter and analyze deep demographic trends and employment rates across different regions.

Objective 5: Ensure database normalization and strict integrity constraints so that data remains clean, connected, and free of duplicates.

4. Scope

What is Included:

- **Geographic & Personnel Infrastructure:** Tracking administrative boundaries from provinces down to local census blocks, along with the operational field staff hierarchies.
- **Property & Household Metrics:** Managing details of physical structures, domestic utility profiles, and family household groupings.
- **Citizen Demographics:** Storing individual citizen records, including identity verification numbers (CNICs), family relations, educational status, and income levels.
- **Backend Database Optimization:** Developing index structures, relational constraints, and core SQL queries for backend data analysis.

What is Not Included:

- **Front-End Applications:** The project does not include building web or mobile applications for data entry or public dashboards.
- **Live GPS Tracking:** It maps static coordinates for structures but does not support real-time GPS tracking of field staff.
- **Biometric Verification System:** The database processes text-based CNIC records but does not handle direct fingerprint or facial biometric verification processing.

5. Entities and Attributes

Entity	Attributes
Water_Source	water_source_id , source_name
Lighting_Source	lighting_source_id , source_name
Cooking_Fuel_Source	fuel_source_id , fuel_source_name
Washroom_Status	washroom_status_id , status_name
Kitchen_Status	kitchen_status_id , status_name
Biological_Sex	sex_id , sex_name
Religion	religion_id , religion_name
Mother_Tongue	mother_tongue_id , tongue_name
Education_Level	education_level_id , level_name
Employment_Status	employment_status_id , status_name
Marital_Status	marital_status_id , status_name
Disability_Type	disability_type_id , disability_name
Occupation	occupation_id , occupation_name
Industry	industry_id , industry_name
Relationship_Type	relationship_type_id , relationship_name

Entity	Attributes
Census_Cycle	census_id , census_year, census_title, start_date, end_date
Census_Staff	staff_id , first_name, last_name, staff_role, contact_number, email, assigned_census_block_id, supervisor_id, employment_date

Entity	Attributes
Nationality	nationality_id , country_name, citizenship_status
Province	province_id , province_name
Division	division_id , division_name, province_id
District	district_id , district_name, division_id
Tehsil	tehsil_id , tehsil_name, district_id
Union_Council	union_council_id , union_council_name, union_council_classification_id, tehsil_id
Census_Block	census_block_id , census_block_code, estimated_household_count, is_urban, area_sq_km, union_council_id
Union_Council_Classification	union_council_classification_id , classification_name, description

Entity	Attributes
Person	person_id , household_id, first_name, last_name, cnic_number, sex_id, date_of_birth, nationality_id, religion_id, mother_tongue_id, marital_status_id, relationship_type_id, head_person_id, is_literate, education_level_id, currently_attending_school, employment_status_id, occupation_id, industry_id, monthly_income, data_entry_operator_id, created_date
Person_Disability	person_disability_id , person_id , disability_type_id
Migration_History	migration_id , person_id , previous_district_id, migration_reason, year_of_migration

Entity	Attributes
Structure_Type	structure_type_id , type_name, description
Structure	structure_id , structure_address_no, gps_latitude, gps_longitude, created_date, census_block_id, structure_type_id, water_source_id, lighting_source_id, fuel_source_id, washroom_status_id, kitchen_status_id
Structure_Residential	structure_id , total_residential_units
Structure_Commercial	structure_id , commercial_type_id, business_name, estimated_active_employees, has_industrial_power_connection, has_active_license
Structure_Institution	structure_id , institution_type, organization_name, population_count, is_government
Commercial_Sector_Type	commercial_type_id , commercial_type_name, description
Household	household_id , structure_id, census_id, household_sub_token, household_category_id, residential_status_type_id, owner_sex_id, number_of_rooms, household_size, data_entry_operator_id, created_date
Household_Category	household_category_id , category_name, description
Residential_Status_Type	residential_status_type_id , status_name, description

6. Relationships

1. Geographic Hierarchy Chain

Province to Division:

- **Cardinality: 1:M** (One Province can have Many Divisions).
- **Participation: Total** on Division side (A division cannot exist without being mapped to a province).
- **Foreign Key:** Division.province_id references Province.province_id.

Division to District:

- **Cardinality: 1:M** (One Division contains Many Districts).
- **Participation: Total** on District side (A district must belong to an administrative division).
- **Foreign Key:** District.division_id references Division.division_id.

District to Tehsil:

- **Cardinality: 1:M** (One District oversees Many Tehsils).
- **Participation: Total** on Tehsil side.
- **Foreign Key:** Tehsil.district_id references District.district_id.

Tehsil to Union Council:

- **Cardinality: 1:M** (One Tehsil contains Many Union Councils).
- **Participation: Total** on Union Council side.
- **Foreign Key:** Union_Council.tehsil_id references Tehsil.tehsil_id.

Union Council to Census Block:

- **Cardinality: 1:M** (One Union Council contains Many localized Census Blocks).
- **Participation: Total** on Census Block side.
- **Foreign Key:** Census_Block.union_council_id references Union_Council.union_council_id.

2. Building Extension Layers

Structure to Structure_Residential:

- **Cardinality: 1:1**
- **Participation: Partial** on Structure side, **Total** on Residential side (The profile cannot exist without its base parent building asset).
- **Foreign Key:** Structure_Residential.structure_id references Structure.structure_id.

Structure to Structure_Commercial:

- **Cardinality: 1:1**
- **Participation: Partial** on Structure side, **Total** on Commercial side.
- **Foreign Key:** Structure_Commercial.structure_id references Structure.structure_id.

Structure to Structure_Institution:

- **Cardinality: 1:1.**
- **Participation: Partial** on Structure side, **Total** on Institution side.
- **Foreign Key:** Structure_Institution.structure_id references Structure.structure_id.

3. Core Operational Census Operations

Census_Block to Structure Location:

- **Cardinality: 1:M** (One Census Block encompasses Many physical building structures).
- **Participation: Total** on Structure side (Every building must be cataloged inside an active census block boundary).
- **Foreign Key:** Structure.census_block_id references Census_Block.census_block_id.

Structure Location to Household Unit:

- **Cardinality: 1:M** (A physical building structure can host Multiple separate living households).
- **Participation: Total** on Household side (Every household must be linked to a physical address/structure asset).
- **Foreign Key:** Household.structure_id references Structure.structure_id.

Household Unit to Person Profile:

- **Cardinality: 1:M** (One Household unit contains Many citizens living/eating together).
- **Participation: Total** on Person side (An individual citizen must be accounted for within an registered household cohort).
- **Foreign Key:** Person.household_id references Household.household_id.

4. Recursive Self-Referencing Links

Census_Staff to Census_Staff (Management Hierarchy):

- **Cardinality: 1:M** (One Supervisor manages Many field staff members/enumerators).
- **Participation: Total** on enumerators side (Enumerators have supervisors).
- **Foreign Key:** Census_Staff.supervisor_id references Census_Staff.staff_id.

Person to Person (Household Head Mapping):

- **Cardinality: 1:M** (One household head is referenced by Many living family members in the house).
- **Participation: Partial** on both sides (A chosen household head points to null or themselves, while family members point to the head).
- **Foreign Key:** Person.head_person_id references Person.person_id.

5. Junction Tables

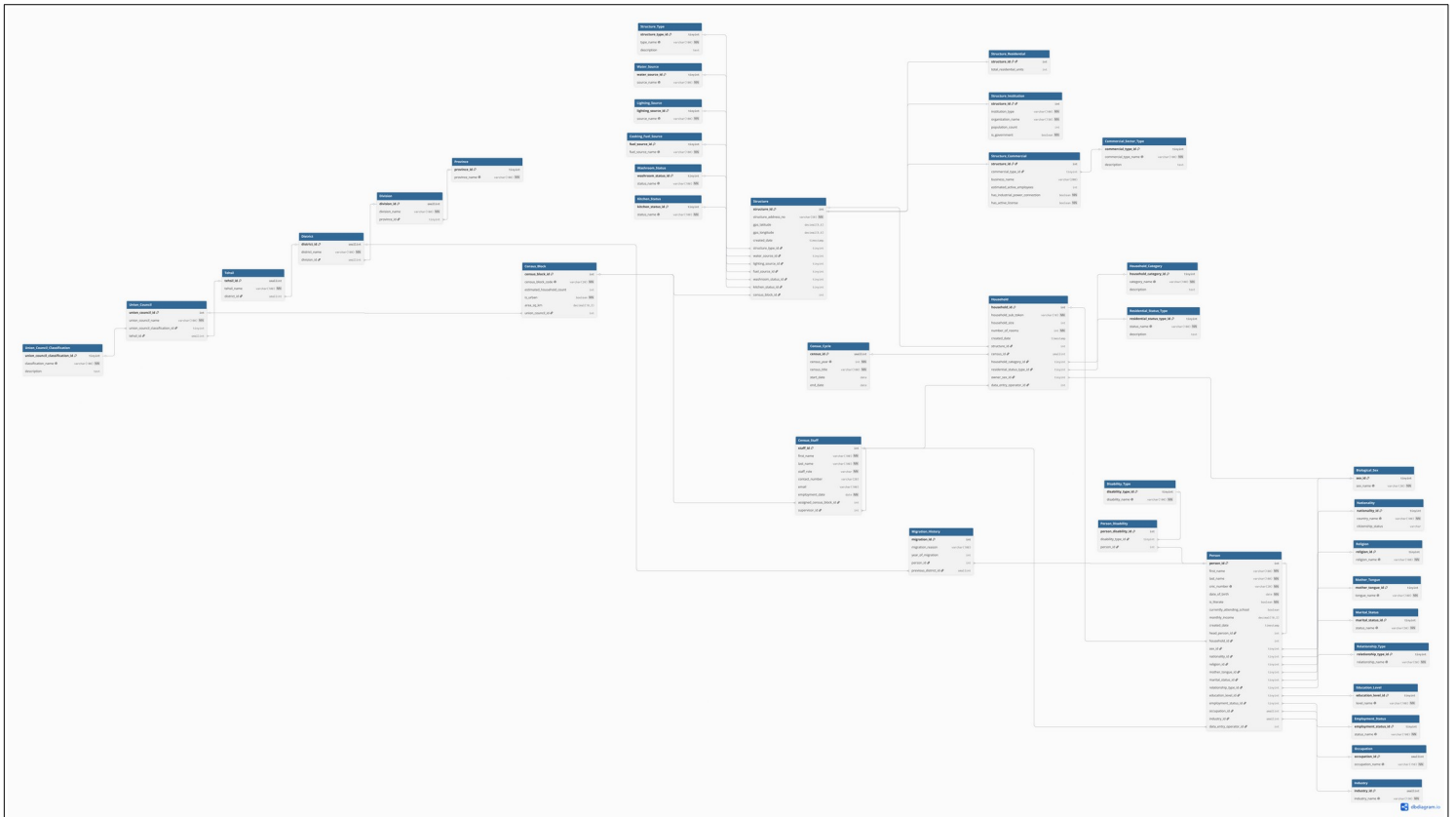
Person to Disability_Type via Person_Disability:

- **Cardinality: M:N** (One person can have multiple concurrent functional health challenges; one disability type applies to many citizens).
- **Participation: Total** on the junction table (Person_Disability), **Partial** on the individual tables (A citizen might have zero disabilities listed).
- **Foreign Keys:** Person_Disability.person_id references Person.person_id. Person_Disability.disability_type_id references Disability_Type.disability_type_id.

Person to District via Migration_History:

- **Cardinality: M:N** (A citizen can move through multiple locations over time; a single district acts as a historical home for many citizens).
- **Participation: Total** on the history tracker (Migration_History), **Partial** on individual tables.
- **Foreign Keys:** Migration_History.person_id references Person.person_id.
Migration_History.previous_district_id references District.district_id.

7. ERD (Entity Relationship Diagram)



8. Relational Schema*

- **Water_Source** (**water_source_id**, **source_name**)
- **Lighting_Source** (**lighting_source_id**, **source_name**)
- **Cooking_Fuel_Source** (**fuel_source_id**, **fuel_source_name**)
- **Washroom_Status** (**washroom_status_id**, **status_name**)
- **Kitchen_Status** (**kitchen_status_id**, **status_name**)
- **Household_Category** (**household_category_id**, **category_name**, **description**)
- **Residential_Status_Type** (**residential_status_type_id**, **status_name**, **description**)
- **Biological_Sex** (**sex_id**, **sex_name**)
- **Religion** (**religion_id**, **religion_name**)
- **Mother_Tongue** (**mother_tongue_id**, **tongue_name**)
- **Education_Level** (**education_level_id**, **level_name**)
- **Employment_Status** (**employment_status_id**, **status_name**)
- **Marital_Status** (**marital_status_id**, **status_name**)
- **Disability_Type** (**disability_type_id**, **disability_name**)
- **Occupation** (**occupation_id**, **occupation_name**)
- **Industry** (**industry_id**, **industry_name**)
- **Relationship_Type** (**relationship_type_id**, **relationship_name**)
- **Nationality** (**nationality_id**, **country_name**, **citizenship_status**)
- **Union_Council_Classification** (**union_council_classification_id**, **classification_name**, **description**)

- **Census_Cycle** (**census_id**, census_year, census_title, start_date, end_date)
- **Structure_Type** (**structure_type_id**, type_name, description)
- **Commercial_Sector_Type** (**commercial_type_id**, commercial_type_name, description)
- **Province** (**province_id**, province_name)
- **Division** (**division_id**, division_name, *province_id references Province.province_id*)
- **District** (**district_id**, district_name, *division_id references Division.division_id*)
- **Tehsil** (**tehsil_id**, tehsil_name, *district_id references District.district_id*)
- **Union_Council** (**union_council_id**, union_council_name, *union_council_classification_id references Union_Council_Classification.union_council_classification_id, tehsil_id references Tehsil.tehsil_id*)
- **Census_Block** (**census_block_id**, census_block_code, estimated_household_count, is_urban, area_sq_km, *union_council_id references Union_Council.union_council_id*)
- **Census_Staff** (**staff_id**, first_name, last_name, staff_role, contact_number, email, *assigned_census_block_id references Census_Block.census_block_id, supervisor_id references Census_Staff.staff_id, employment_date*)
- **Structure** (**structure_id**, structure_address_no, gps_latitude, gps_longitude, created_date, *census_block_id references Census_Block.census_block_id, structure_type_id references Structure_Type.structure_type_id, water_source_id references Water_Source.water_source_id, lighting_source_id references Lighting_Source.lighting_source_id, fuel_source_id references Cooking_Fuel_Source.fuel_source_id, washroom_status_id references Washroom_Status.washroom_status_id, kitchen_status_id references Kitchen_Status.kitchen_status_id*)
- **Structure_Institution** (**structure_id** *references Structure.structure_id*, institution_type, organization_name, population_count, is_government)

- **Structure_Commercial** (**structure_id** references *Structure.structure_id*, *commercial_type_id* references *Commercial_Sector_Type.commercial_type_id*, *business_name*, *estimated_active_employees*, *has_industrial_power_connection*, *has_active_license*)
- **Structure_Residential** (**structure_id** references *Structure.structure_id*, *total_residential_units*)
- **Household** (**household_id**, *structure_id* references *Structure.structure_id*, *census_id* references *Census_Cycle.census_id*, *household_sub_token*, *household_category_id* references *Household_Category.household_category_id*, *residential_status_type_id* references *Residential_Status_Type.residential_status_type_id*, *owner_sex_id* references *Biological_Sex.sex_id*, *number_of_rooms*, *household_size*, *data_entry_operator_id* references *Census_Staff.staff_id*, *created_date*)
- **Person** (**person_id**, *household_id* references *Household.household_id*, *first_name*, *last_name*, *cnic_number*, *sex_id* references *Biological_Sex.sex_id*, *date_of_birth*, *nationality_id* references *Nationality.nationality_id*, *religion_id* references *Religion.religion_id*, *mother_tongue_id* references *Mother_Tongue.mother_tongue_id*, *marital_status_id* references *Marital_Status.marital_status_id*, *relationship_type_id* references *Relationship_Type.relationship_type_id*, *head_person_id* references *Person.person_id*, *is_literate*, *education_level_id* references *Education_Level.education_level_id*, *currently_attending_school*, *employment_status_id* references *Employment_Status.employment_status_id*, *occupation_id* references *Occupation.occupation_id*, *industry_id* references *Industry.industry_id*, *monthly_income*, *data_entry_operator_id* references *Census_Staff.staff_id*, *created_date*)
- **Person_Disability** (**person_disability_id**, *person_id* references *Person.person_id*, *disability_type_id* references *Disability_Type.disability_type_id*)
- **Migration_History** (**migration_id**, *person_id* references *Person.person_id*, *previous_district_id* references *District.district_id*, *migration_reason*, *year_of_migration*)

***HOW TO READ**

- Fields styled in **bold** represent the **Primary Key (PK)** of the table.
- Fields styled in *italics* represent a **Foreign Key (FK)** along with its explicit relational table direction map.
- Fields styled in ***bold italics*** represent a composite setup acting concurrently as both the table's Primary Key and its parent class Foreign Key identifier (1:1 structural extension).

9. SQL Queries and Data Retrieval

A. Joins

INNER JOIN

Shows top 5 districts in Pakistan with highest number of households.

```
mysql> SELECT
->     p.province_name,
->     d.district_name,
->     SUM(cb.estimated_household_count) AS total_population
-> FROM Province p
-> INNER JOIN Division dv ON p.province_id = dv.province_id
-> INNER JOIN District d ON dv.division_id = d.division_id
-> INNER JOIN Tehsil t ON d.district_id = t.district_id
-> INNER JOIN Union_Council uc ON t.tehsil_id = uc.tehsil_id
-> INNER JOIN Census_Block cb ON uc.union_council_id = cb.union_council_id
-> GROUP BY p.province_name, d.district_name
-> ORDER BY Total_Population DESC
-> LIMIT 5;
```

province_name	district_name	total_population
Sindh	Karachi Central	3440
Sindh	Karachi South	3350
Islamabad Capital Territory	Islamabad Urban	2800
Sindh	Karachi East	2650
Sindh	Hyderabad	1910

5 rows in set (0.01 sec)

Shows people who have Bachelors, Masters or PhD degree with their job and income.

```
mysql> SELECT
->     CONCAT(p.first_name, ' ', p.last_name) AS name,
->     el.level_name,
->     es.status_name,
->     o.occupation_name,
->     i.industry_name,
->     p.monthly_income
-> FROM Person p
-> INNER JOIN Education_Level el ON p.education_level_id = el.education_level_id
-> INNER JOIN Employment_Status es ON p.employment_status_id = es.employment_status_id
-> INNER JOIN Occupation o ON p.occupation_id = o.occupation_id
-> INNER JOIN Industry i ON p.industry_id = i.industry_id
-> WHERE UPPER(el.level_name) IN ('BACHELORS', 'MASTERS', 'PHD');
```

name	level_name	status_name	occupation_name	industry_name	monthly_income
Muhammad Khan	Masters	Employed Full-Time	Doctor/Medical Professional	Public Administration	45000.00
Ali Khan	Masters	Student	Government Official	Wholesale and Retail Trade	0.00
Malik Hassan	PHD	Employed Full-Time	Primary School Teacher	Utilities (Water/Electricity)	35000.00
Khalid Ahmed	Bachelors	Employed Full-Time	Tailor/Seamstress	Transportation and Logistics	55000.00

4 rows in set (0.00 sec)

LEFT JOIN

Shows all persons with their education, job, occupation and nationality — even if some info is missing.

```
mysql> SELECT
->   CONCAT(p.first_name, ' ', p.last_name) AS name,
->   el.level_name,
->   es.status_name AS employment_status,
->   o.occupation_name,
->   i.industry_name,
->   n.country_name
-> FROM Person p
-> LEFT JOIN Education_Level el ON p.education_level_id = el.education_level_id
-> LEFT JOIN Employment_Status es ON p.employment_status_id = es.employment_status_id
-> LEFT JOIN Occupation o ON p.occupation_id = o.occupation_id
-> LEFT JOIN Industry i ON p.industry_id = i.industry_id
-> LEFT JOIN Nationality n ON p.nationality_id = n.nationality_id
-> ORDER BY el.level_name;
```

name	level_name	employment_status	occupation_name	industry_name	country_name
Khalid Ahmed	Bachelors	Employed Full-Time	Tailor/Seamstress	Transportation and Logistics	Pakistan
Hamza Hassan	Intermediate	Employed Full-Time	Carpenter	Telecommunications	Pakistan
Danial Khan	Intermediate	Employed Full-Time	Construction Laborer	Public Administration	Pakistan
Ayesha Khan	Intermediate	Student	NULL	NULL	Pakistan
Kamil Ahmad	Intermediate	Employed Full-Time	Plumber	Manufacturing - Other	Pakistan
Taufiq Malik	Intermediate	Employed Full-Time	Engineer	Transportation and Logistics	Pakistan
Rohan Hassan	Intermediate	Employed Full-Time	Electrician	Public Administration	Pakistan
Karim Ali	Intermediate	Employed Full-Time	Shopkeeper/Retail Merchant	Health and Social Work	Pakistan
Mir Khan	Intermediate	Employed Full-Time	Nurse	Construction	Pakistan
Fachan Hassan	Intermediate	Employed Full-Time	Driver/Transport Worker	Education	Pakistan
Dina Ahmed	Intermediate	Employed Full-Time	NULL	NULL	Pakistan
Bashir Khan	Intermediate	Student	Engineer	Wholesale and Retail Trade	Pakistan
Abbas Malik	Intermediate	Employed Full-Time	Tailor/Seamstress	Manufacturing - Textiles	Pakistan
Muhammad Khan	Masters	Employed Full-Time	Doctor/Medical Professional	Public Administration	Pakistan
Ali Khan	Masters	Student	Government Official	Wholesale and Retail Trade	Pakistan
Imad Ali	Matriculation	Student	NULL	NULL	Pakistan
Ahmed Ali	Matriculation	Employed Full-Time	Driver/Transport Worker	Agriculture and Farming	Pakistan
Arjun Ali	Matriculation	Student	NULL	NULL	Pakistan
Akhtar Hassan	Matriculation	Employed Full-Time	Business Owner	Hospitality and Tourism	Pakistan
Mansoor Hassan	Matriculation	Employed Full-Time	Driver/Transport Worker	Public Administration	Pakistan
Zubair Hassan	Matriculation	Employed Full-Time	Factory Worker	Health and Social Work	Pakistan
Wasim Malik	Matriculation	Employed Full-Time	Agricultural Farmer	Manufacturing - Textiles	Pakistan
Rana Ahmed	Matriculation	Employed Full-Time	Domestic Helper/Maid	Health and Social Work	Pakistan
Manzoor Ahmad	Matriculation	Employed Full-Time	Construction Laborer	Wholesale and Retail Trade	Pakistan
Parah Khan	Matriculation	Employed Full-Time	NULL	NULL	Pakistan
Nadim Khan	Matriculation	Employed Full-Time	Government Official	Arts and Entertainment	Pakistan
Rasheed Ali	Matriculation	Employed Full-Time	Nurse	Wholesale and Retail Trade	Pakistan

Shows all married persons with their household size information.

```
mysql> SELECT
->   CONCAT(p.first_name, ' ', p.last_name) AS name,
->   ms.status_name,
->   h.household_size
-> FROM Person p
-> LEFT JOIN Marital_Status ms ON p.marital_status_id = ms.marital_status_id
-> LEFT JOIN Household h ON p.household_id = h.household_id
-> WHERE ms.status_name = 'Married';
```

name	status_name	household_size
Muhammad Khan	Married	5
Fatima Khan	Married	5
Ahmed Ali	Married	4
Noor Ali	Married	4
Rashid Hassan	Married	4
Zainab Hassan	Married	4
Nasir Hussain	Married	4
Amina Hussain	Married	4
Gamir Khan	Married	3
Maha Khan	Married	3
Waqar Ahmed	Married	4
Rania Ahmed	Married	4
Jamal Malik	Married	3
Lila Malik	Married	3
Tariq Hassan	Married	3
Nadia Hassan	Married	3
Yousef Ali	Married	3
Hana Ali	Married	3
Saad Khan	Married	3
Luna Khan	Married	3
Adnan Ahmed	Married	3
Yasmin Ahmed	Married	3
Wasim Malik	Married	3
Rena Malik	Married	3
Mohan Hassan	Married	3
Anjali Hassan	Married	3
Deepak Ali	Married	3
Priya Ali	Married	3
Siddiqui Khan	Married	3
Gulnar Khan	Married	3
Zahir Ahmed	Married	3
Meena Ahmed	Married	3
Ibrahim Raza	Married	3
Saira Raza	Married	3

RIGHT JOIN

Shows urban census blocks with more than 300 households and their assigned staff.

```
mysql> SELECT
->   CONCAT(cs.first_name, ' ', cs.last_name) AS staff_name,
->   cs.staff_role,
->   cb.contact_number,
->   cb.census_block_code,
->   cb.estimated_household_count,
->   cb.area_sq_km,
->   uc.union_council_name,
->   t.tehsil_name,
->   d.district_name
-> FROM Census_Staff cs
-> RIGHT JOIN Census_Block cb ON cs.assigned_census_block_id = cb.census_block_id
-> RIGHT JOIN Union_Council uc ON cb.union_council_id = uc.union_council_id
-> RIGHT JOIN Tehsil t ON uc.tehsil_id = t.tehsil_id
-> RIGHT JOIN District d ON t.district_id = d.district_id
-> WHERE cb.is_urban = TRUE AND cb.estimated_household_count > 300 AND (cs.staff_role IN ('Enumerator', 'Supervisor') OR cs.staff_role IS NULL)
-> ORDER BY cb.estimated_household_count DESC;
```

staff_name	staff_role	contact_number	census_block_code	estimated_household_count	area_sq_km	union_council_name	tehsil_name	district_name
NULL	NULL	NULL	BLK-UC062	930	2.10	UC Lyari	Saddar (KHI South)	Karachi South
NULL	NULL	NULL	BLK-UC070	920	1.70	UC Azizabad	Gulberg (KHI)	Karachi Central
NULL	NULL	NULL	BLK-UC069	880	1.90	UC Federal B Area	Gulberg (KHI)	Karachi Central
NULL	NULL	NULL	BLK-UC064	850	1.50	UC Kharadar	Aram Bagh	Karachi South
NULL	NULL	NULL	BLK-UC072	830	1.40	UC Nazimabad	Liaquatabad	Karachi Central
NULL	NULL	NULL	BLK-UC071	810	1.60	UC Liaquatabad Central	Liaquatabad	Karachi Central
NULL	NULL	NULL	BLK-UC061	800	4.50	UC Clifton	Saddar (KHI South)	Karachi South
NULL	NULL	NULL	BLK-UC063	750	1.40	UC Saddar KHI	Aram Bagh	Karachi South
NULL	NULL	NULL	BLK-UC193	750	2.00	UC F-7	F-Series Sectors	Islamabad Urban
NULL	NULL	NULL	BLK-UC194	720	2.20	UC F-8	F-Series Sectors	Islamabad Urban
NULL	NULL	NULL	BLK-UC066	710	2.40	UC Ferozabad	Gulshan-e-Johal	Karachi East
NULL	NULL	NULL	BLK-UC065	680	3.50	UC Gulshan Central	Gulshan-e-Johal	Karachi East
NULL	NULL	NULL	BLK-UC195	680	2.50	UC G-9	G-Series Sectors	Islamabad Urban
NULL	NULL	NULL	BLK-UC196	650	2.80	UC G-10	G-Series Sectors	Islamabad Urban
NULL	NULL	NULL	BLK-UC068	640	2.40	UC Tariq Road	Ferozabad	Karachi East
NULL	NULL	NULL	BLK-UC067	620	2.50	UC PECHS	Ferozabad	Karachi East
NULL	NULL	NULL	BLK-UC197	620	2.20	UC Nawa Kilili	Quetta City	Quetta
NULL	NULL	NULL	BLK-UC117	600	6.50	UC Nowshera Cantt	Nowshera City	Nowshera
NULL	NULL	NULL	BLK-UC138	600	2.40	UC Satellite Town QTA	Quetta City	Quetta
NULL	NULL	NULL	BLK-UC109	580	3.50	UC Hayatabad	Peshawar City	Peshawar
NULL	NULL	NULL	BLK-UC118	580	7.00	UC Rinalpur	Nowshera City	Nowshera
NULL	NULL	NULL	BLK-UC132	580	3.50	UC Tarbela Dam Sector	Topi	Swabi
NULL	NULL	NULL	BLK-UC143	570	2.50	UC Mardan Central	Mardan City	Mardan
NULL	NULL	NULL	BLK-UC110	560	3.40	UC University Town	Peshawar City	Peshawar
Buashra Ansari	Enumerator	0300-1000035	BLK-UC032	550	6.20	UC Shorkot Cantt	Shorkot	Jhang
NULL	NULL	NULL	BLK-UC082	550	5.20	UC SITE Kotri	Kotri	Jamshoro
NULL	NULL	NULL	BLK-UC146	550	2.70	UC Bori	Mardan City	Mardan
NULL	NULL	NULL	BLK-UC137	540	2.40	UC Mansehra Central	Mansehra City	Mansehra
NULL	NULL	NULL	BLK-UC097	530	2.40	UC Larkana Central	Larkana City	Larkana
NULL	NULL	NULL	BLK-UC073	520	3.50	UC Latifabad	Hyderabad City	Hyderabad
NULL	NULL	NULL	BLK-UC144	520	6.00	UC Tarbela	Chari	Haripur
NULL	NULL	NULL	BLK-UC098	510	2.40	UC Darsi	Larkana City	Larkana
NULL	NULL	NULL	BLK-UC121	510	2.50	UC Mingora City	Babozai (Mingora)	Swat

Shows educated and full-time employed persons with their religion and mother tongue profile.

```
mysql> SELECT
->   CONCAT(p.first_name, ' ', p.last_name) AS name,
->   p.religion_name,
->   mt.tongue_name AS mother_tongue,
->   el.level_name AS education_level,
->   es.status_name AS employment_status,
->   p.monthly_income,
->   TIMEDIFF(YEAR, p.date_of_birth, CURDATE()) AS age
-> FROM Religion r
-> RIGHT JOIN Person p ON r.religion_id = p.religion_id
-> RIGHT JOIN Household h ON p.household_id = h.household_id
-> RIGHT JOIN Mother_Tongue mt ON p.mother_tongue_id = mt.mother_tongue_id
-> RIGHT JOIN Education_Level el ON p.education_level_id = el.education_level_id
-> LEFT JOIN Employment_Status es ON p.employment_status_id = es.employment_status_id
-> WHERE UPPER(el.level_name) IN ('INTERMEDIATE', 'BACHELORS', 'MASTERS', 'PHD') AND es.status_name = 'Employed Full-Time' AND p.monthly_income > 30000
-> ORDER BY p.monthly_income DESC;
```

name	religion_name	mother_tongue	education_level	employment_status	monthly_income	age
Rhaid Ahmed	Islam	Urdu	Bachelors	Employed Full-Time	55000.00	40
Muhammad Khan	Islam	Urdu	Masters	Employed Full-Time	45000.00	61
Malik Hassan	Islam	Punjabi	PHD	Employed Full-Time	35000.00	33
Hamza Hassan	Christianity	Hindko	Intermediate	Employed Full-Time	34000.00	34
Kamil Ahmad	Islam	Urdu	Intermediate	Employed Full-Time	33000.00	30
Mir Khan	Islam	Bataiki	Intermediate	Employed Full-Time	32000.00	32
Karim Ali	Islam	Urdu	Intermediate	Employed Full-Time	32000.00	32
Farhan Hassan	Islam	Urdu	Intermediate	Employed Full-Time	31000.00	32

0 rows in set (0.01 sec)

NATURAL JOIN

Shows married individuals with an income above 20,000, joined implicitly on marital_status_id.

```
mysql> SELECT
->   p.first_name,
->   p.last_name,
->   p.date_of_birth,
->   p.monthly_income,
->   ms.status_name
-> FROM Marital_Status ms
-> NATURAL JOIN (
->   SELECT marital_status_id, first_name, last_name, date_of_birth, monthly_income
->   FROM Person
-> ) p
-> WHERE ms.status_name = 'Married' AND p.monthly_income > 20000
-> ORDER BY p.monthly_income DESC;
```

first_name	last_name	date_of_birth	monthly_income	status_name
Yousef	Ali	1958-01-08	72000.00	Married
Rashad	Ahmed	1959-09-19	71000.00	Married
Faisal	Hassan	1956-12-17	68000.00	Married
Waqar	Ahmed	1955-12-10	65000.00	Married
Zahir	Ahmad	1960-06-14	62000.00	Married
Shabir	Khan	1958-01-25	60000.00	Married
Mohan	Hassan	1957-05-12	58000.00	Married
Riaz	Ali	1958-07-15	55000.00	Married
Nadim	Khan	1968-09-02	53000.00	Married
Samir	Khan	1970-04-20	52000.00	Married
Zubair	Hassan	1966-03-14	51000.00	Married
Wasim	Malik	1969-07-09	51000.00	Married
Akhtar	Hassan	1963-04-08	50000.00	Married
Habib	Malik	1967-03-06	49000.00	Married
Nasir	Hussain	1962-11-05	48000.00	Married
Manzoor	Ahmad	1969-05-21	48000.00	Married
Salim	Ahmed	1965-02-09	47000.00	Married
Nawaz	Ali	1964-07-06	46000.00	Married
Ahnan	Ahmed	1961-10-25	46000.00	Married
Muhammad	Khan	1965-03-15	45000.00	Married
Deepak	Ali	1964-02-22	44000.00	Married
Feroz	Malik	1961-11-20	44000.00	Married
Babar	Khan	1962-11-28	43000.00	Married
Rashid	Hassan	1960-01-22	42000.00	Married
Jalal	Khan	1962-08-10	42000.00	Married
Jamal	Malik	1972-06-18	41000.00	Married
Ibrahim	Raza	1959-08-03	40000.00	Married
Qasim	Malik	1960-10-11	39000.00	Married
Tariq	Hassan	1963-08-29	39000.00	Married
Ahmed	Ali	1968-07-19	38000.00	Married
Saad	Khan	1966-04-17	37000.00	Married
Siddiqui	Khan	1961-09-30	35000.00	Married

32 rows in set (0.00 sec)

Shows individuals with premium employment statuses earning over 30,000, joined implicitly on employment_status_id.

```
mysql> SELECT
->   p.first_name,
->   p.last_name,
->   p.monthly_income,
->   p.date_of_birth,
->   es.status_name
-> FROM Employment_Status es
-> NATURAL JOIN (
->   SELECT employment_status_id, first_name, last_name, monthly_income, date_of_birth
->   FROM Person
-> ) p
-> WHERE es.status_name IN ('Employed Full-Time', 'Self-Employed') AND p.monthly_income > 30000
-> ORDER BY p.monthly_income DESC;
```

first_name	last_name	monthly_income	date_of_birth	status_name
Rashad	Ahmed	71000.00	1959-09-19	Employed Full-Time
Faisal	Hassan	68000.00	1956-12-17	Employed Full-Time
Zahir	Ahmad	62000.00	1960-06-14	Employed Full-Time
Shabir	Khan	60000.00	1958-01-25	Employed Full-Time
Mohan	Hassan	58000.00	1957-05-12	Employed Full-Time
Riaz	Ali	55000.00	1958-07-15	Employed Full-Time
Khalid	Ahmed	55000.00	1955-09-13	Employed Full-Time
Nadim	Khan	53000.00	1968-09-02	Employed Full-Time
Samir	Khan	52000.00	1970-04-20	Employed Full-Time
Zubair	Hassan	51000.00	1966-03-14	Employed Full-Time
Wasim	Malik	51000.00	1969-07-09	Employed Full-Time
Akhtar	Hassan	50000.00	1963-04-08	Employed Full-Time
Habib	Malik	49000.00	1967-03-06	Employed Full-Time
Nasir	Hussain	48000.00	1962-11-05	Employed Full-Time
Manzoor	Ahmad	48000.00	1969-05-21	Employed Full-Time
Salim	Ahmed	47000.00	1965-02-09	Employed Full-Time
Nawaz	Ali	46000.00	1964-07-06	Employed Full-Time
Ahnan	Ahmed	46000.00	1961-10-25	Employed Full-Time
Muhammad	Khan	45000.00	1965-03-15	Employed Full-Time
Feroz	Malik	44000.00	1961-11-20	Employed Full-Time
Deepak	Ali	44000.00	1964-02-22	Employed Full-Time
Babar	Khan	43000.00	1962-11-28	Employed Full-Time
Rashid	Hassan	42000.00	1960-01-22	Employed Full-Time
Jalal	Khan	42000.00	1962-08-10	Employed Full-Time
Jamal	Malik	41000.00	1972-06-18	Employed Full-Time
Ibrahim	Raza	40000.00	1959-08-03	Employed Full-Time
Tariq	Hassan	39000.00	1963-08-29	Employed Full-Time
Qasim	Malik	39000.00	1960-10-11	Employed Full-Time
Ahmed	Ali	38000.00	1968-07-19	Employed Full-Time
Saad	Khan	37000.00	1966-04-17	Employed Full-Time
Ashar	Ahmed	36000.00	1963-06-22	Employed Full-Time
Siddiqui	Khan	35000.00	1961-09-30	Employed Full-Time
Malik	Hassan	35000.00	1992-10-30	Employed Full-Time
Amna	Hassan	34000.00	1991-10-22	Employed Full-Time
Kamil	Ahmad	33000.00	1996-02-10	Employed Full-Time
Mir	Khan	32000.00	1994-03-29	Employed Full-Time
Karim	Ali	32000.00	1993-12-05	Employed Full-Time

SELF JOIN

Maps out the internal management hierarchy of the census field staff team.

```
mysql> SELECT
->   CONCAT(emp.first_name, ' ', emp.last_name) AS staff_member,
->   emp.staff_role AS Staff_Role,
->   CONCAT(mgr.first_name, ' ', mgr.last_name) AS direct_supervisor
-> FROM Census_Staff emp
-> INNER JOIN Census_Staff mgr ON emp.supervisor_id = mgr.staff_id;
```

staff_member	Staff_Role	direct_supervisor
Ali Hassan	Enumerator	Ahmed Raza
Zainab Noor	Enumerator	Ahmed Raza
Bilal Khan	Enumerator	Fatima Bibi
Maria Siddiqui	Enumerator	Fatima Bibi
Omar Farooq	Enumerator	Tariq Mehmood
Sana Javed	Enumerator	Tariq Mehmood
Iman Ghani	Enumerator	Ahmed Raza
Hina Nae	Enumerator	Ahmed Raza
Imran Shah	Enumerator	Fatima Bibi
Sadaf Ali	Enumerator	Fatima Bibi
Hanza Rehman	Enumerator	Tariq Mehmood
Aqsa Iqbal	Enumerator	Tariq Mehmood
Rashid Latif	Data_Entry_Operator	Ahmed Raza
Rabia Basri	Data_Entry_Operator	Ahmed Raza
Waqas Ahmed	Data_Entry_Operator	Fatima Bibi
Nadia Gul	Data_Entry_Operator	Fatima Bibi
Zeeshan Malik	Data_Entry_Operator	Tariq Mehmood
Anna Farveem	Data_Entry_Operator	Tariq Mehmood
Salman Farsi	Quality_Checker	Ahmed Raza
Sadia Inran	Quality_Checker	Ahmed Raza
Faizan Aslam	Quality_Checker	Fatima Bibi
Sidra Munir	Quality_Checker	Fatima Bibi
Adnan Sami	Quality_Checker	Tariq Mehmood
Madiha Khalid	Quality_Checker	Tariq Mehmood
Shahid Afidi	Enumerator	Ahmed Raza
Mehwish Hayat	Enumerator	Ahmed Raza
Daniyal Arshad	Enumerator	Fatima Bibi
Arooj Fatima	Enumerator	Fatima Bibi
Bilal Saeed	Enumerator	Tariq Mehmood
Anum Zehra	Enumerator	Tariq Mehmood
Naveed Ul-Haq	Enumerator	Ahmed Raza
Bushra Ansari	Enumerator	Ahmed Raza
Junaid Janshied	Enumerator	Fatima Bibi
Sumaira Khan	Enumerator	Fatima Bibi
Fahad Mustafa	Enumerator	Tariq Mehmood
Mariyam Nawaz	Enumerator	Tariq Mehmood
Saqib Mehmood	Enumerator	Ahmed Raza
Saba Qamar	Enumerator	Ahmed Raza
Waseem Akram	Enumerator	Fatima Bibi
Sofia Khan	Enumerator	Fatima Bibi
Yasir Shah	Enumerator	Tariq Mehmood
Ambar Bashir	Enumerator	Tariq Mehmood
Zahid Iqbal	Enumerator	Ahmed Raza
Shadia Farveem	Enumerator	Ahmed Raza

Maps out biological family dynamics by identifying the official household head for each member.

```
mysql> SELECT
->   CONCAT(member.first_name, ' ', member.last_name) AS family_member,
->   member.cnrc_number AS Member_CNRC,
->   CONCAT(head.first_name, ' ', head.last_name) AS household_head,
->   head.cnrc_number AS Head_CNRC
-> FROM Person member
-> INNER JOIN Person head ON member.head_person_id = head.person_id;
```

family_member	Member_CNRC	household_head	Head_CNRC
Fatima Khan	12345-6000002-3	Muhammad Khan	12345-6000001-1
Ali Khan	12345-6000003-5	Muhammad Khan	12345-6000001-1
Aysha Khan	12345-6000004-7	Muhammad Khan	12345-6000001-1
Hasan Khan	12345-6000005-9	Muhammad Khan	12345-6000001-1
Noor Ali	12345-6000007-2	Ahmed Ali	12345-6000006-0
Karim Ali	12345-6000008-4	Ahmed Ali	12345-6000006-0
Sara Ali	12345-6000009-6	Ahmed Ali	12345-6000006-0
Zainab Hassan	12345-6000011-0	Rashid Hassan	12345-6000010-8
Malik Hassan	12345-6000012-2	Rashid Hassan	12345-6000010-8
Hina Hassan	12345-6000013-4	Rashid Hassan	12345-6000010-8
Amna Hussain	12345-6000015-8	Nasir Hussain	12345-6000014-6
Taran Hussain	12345-6000016-1	Nasir Hussain	12345-6000014-6
Leila Hussain	12345-6000017-3	Nasir Hussain	12345-6000014-6
Maha Khan	12345-6000019-7	Samir Khan	12345-6000018-5
Rashid Khan	12345-6000020-9	Samir Khan	12345-6000018-5
Rania Ahmed	12345-6000022-2	Waqar Ahmed	12345-6000021-0
Khalid Ahmed	12345-6000023-4	Waqar Ahmed	12345-6000021-0
Dina Ahmed	12345-6000024-6	Waqar Ahmed	12345-6000021-0
Ula Malik	12345-6000026-1	Jamal Malik	12345-6000025-8
Amir Malik	12345-6000027-3	Jamal Malik	12345-6000025-8
Nadia Hassan	12345-6000029-7	Tariq Hassan	12345-6000028-5
Faizan Hassan	12345-6000030-9	Tariq Hassan	12345-6000028-5
Hana Ali	12345-6000032-2	Yousef Ali	12345-6000031-0
Ranbeer Ali	12345-6000033-4	Yousef Ali	12345-6000031-0
Luna Khan	12345-6000035-8	Saad Khan	12345-6000034-6
Aziz Khan	12345-6000036-1	Saad Khan	12345-6000034-6
Yasmin Ahmed	12345-6000038-5	Adnan Ahmed	12345-6000037-3
Rana Ahmed	12345-6000039-7	Adnan Ahmed	12345-6000037-3
Rena Malik	12345-6000041-0	Wasim Malik	12345-6000040-9
Raees Malik	12345-6000042-2	Wasim Malik	12345-6000040-9
Anjali Hassan	12345-6000044-6	Mohan Hassan	12345-6000043-4
Rohan Hassan	12345-6000045-8	Mohan Hassan	12345-6000043-4
Pritya Ali	12345-6000047-3	Deepak Ali	12345-6000046-1
Arian Ali	12345-6000048-5	Deepak Ali	12345-6000046-1
Gulnar Khan	12345-6000050-9	Siddiqui Khan	12345-6000049-7
Farah Khan	12345-6000051-0	Siddiqui Khan	12345-6000049-7
Merveen Ahmed	12345-6000053-4	Zahir Ahmed	12345-6000052-2
Kamil Ahmed	12345-6000054-6	Zahir Ahmed	12345-6000052-2
Saira Raza	12345-6000056-1	Ibrahim Raza	12345-6000055-8
Konban Raza	12345-6000057-3	Ibrahim Raza	12345-6000055-8
Marian Khan	12345-6000059-7	Babar Khan	12345-6000058-5

B. Nested and Correlated Queries

Retrieves names of citizens who live in crowded families with more than 4 members.

```
mysql> SELECT first_name, last_name
-> FROM Person
-> WHERE household_id IN (
->     SELECT household_id
->     FROM Household
->     WHERE household_size > 4
-> );
```

first_name	last_name
Muhammad	Khan
Fatima	Khan
Ali	Khan
Ayesha	Khan
Hassan	Khan

5 rows in set (0.00 sec)

Finds citizens earning more than at least one individual of biological sex male.

```
mysql> SELECT first_name, monthly_income
-> FROM Person
-> WHERE monthly_income > ANY (
->     SELECT monthly_income
->     FROM Person
->     WHERE sex_id = 2
-> );
```

first_name	monthly_income
Muhammad	45000.00
Ahmed	38000.00
Karim	32000.00
Rashid	42000.00
Malik	35000.00
Nasir	48000.00
Imran	28000.00
Samir	52000.00
Waqar	65000.00
Khalid	55000.00
Dina	25000.00
Jamal	41000.00
Tariq	39000.00
Farhan	31000.00
Yousef	72000.00
Rasheed	29000.00
Saad	37000.00
Adnan	46000.00
Rana	24000.00
Wasim	51000.00
Mohan	58000.00
Rohan	26000.00
Deepak	44000.00
Siddiqui	35000.00
Farah	18000.00
Zahir	62000.00
Kamil	33000.00
Ibrahim	40000.00
Kashan	23000.00
Babar	43000.00
Danial	27000.00
Habib	49000.00
Faisal	68000.00

Finds citizens earning more than all females

```
mysql> SELECT first_name, monthly_income
-> FROM Person
-> WHERE monthly_income > ALL (
->   SELECT monthly_income
->     FROM Person
->   WHERE sex_id = 2 AND monthly_income IS NOT NULL
-> );
```

first_name	monthly_income
Muhammad	45000.00
Ahmed	38000.00
Karim	32000.00
Rashid	42000.00
Malik	35000.00
Nasir	48000.00
Imran	28000.00
Samir	52000.00
Waqar	65000.00
Khalid	55000.00
Jamal	41000.00
Tariq	39000.00
Farhan	31000.00
Yousef	72000.00
Rasheed	29000.00
Saad	37000.00
Adnan	46000.00
Wasim	51000.00
Mohan	58000.00
Rohan	26000.00
Deepak	44000.00
Siddiqui	35000.00
Zahir	62000.00
Kamil	33000.00
Ibrahim	40000.00
Babar	43000.00
Danial	27000.00
Habib	49000.00
Faisal	68000.00
Hamza	34000.00
Salim	47000.00
Riaz	55000.00
Nadim	53000.00
Qasim	39000.00

Lists all citizens who have never migrated or shifted their residential district.

```
mysql> SELECT first_name, last_name
-> FROM Person
-> WHERE person_id NOT IN (
->   SELECT person_id
->     FROM Migration_History
-> );
```

first_name	last_name
Farah	Khan
Zahir	Ahmad
Meena	Ahmad
Kamil	Ahmad
Ibrahim	Raza
Saira	Raza
Kashan	Raza
Babar	Khan
Mariam	Khan
Danial	Khan
Habib	Malik
Zara	Malik
Karim	Malik
Faisal	Hassan
Iqra	Hassan
Hamza	Hassan
Salim	Ahmed
Saba	Ahmed
Hassam	Ahmed
Riaz	Ali
Bushra	Ali
Imad	Ali
Nadim	Khan
Suhana	Khan
Shiraz	Khan
Qasim	Malik
Hafsa	Malik

Finds citizens earning less than or equal to the absolute lowest salary among males.

```
mysql> SELECT first_name, last_name
-> FROM Person
-> WHERE monthly_income <= ALL (
->   SELECT monthly_income
->   FROM Person
->   WHERE sex_id = 1
-> );
```

first_name	last_name
Ali	Khan
Ayesha	Khan
Hassan	Khan
Sara	Ali
Hina	Hassan
Leila	Hussain
Bashir	Khan
Amir	Malik
Aziz	Khan
Naeem	Malik
Arjun	Ali
Karim	Malik
Hassam	Ahmed
Imad	Ali
Shiraz	Khan
Sarmad	Ahmad
Rizwan	Ali
Suleman	Hassan
Shafiq	Khan

19 rows in set (0.00 sec)

Finds citizens earning less than or equal to at least one individual of among males.

```
mysql> SELECT first_name, last_name
-> FROM Person
-> WHERE monthly_income <= ANY (
->   SELECT monthly_income
->   FROM Person
->   WHERE sex_id = 1
-> );
```

first_name	last_name
Muhammad	Khan
Ali	Khan
Ayesha	Khan
Hassan	Khan
Ahmed	Ali
Karim	Ali
Sara	Ali
Rashid	Hassan
Malik	Hassan
Hina	Hassan
Nasir	Hussain
Imran	Hussain
Leila	Hussain
Samir	Khan
Bashir	Khan
Waqar	Ahmed
Khalid	Ahmed
Dina	Ahmed
Jamal	Malik
Amir	Malik
Tariq	Hassan
Farhan	Hassan
Yousef	Ali
Rasheed	Ali
Saad	Khan
Aziz	Khan
Adnan	Ahmed

Identifies citizens who have an active internal migration or relocation history record.

```
mysql> SELECT p.first_name, p.last_name
-> FROM Person p
-> WHERE EXISTS (
->     SELECT 1
->     FROM Migration_History m
->     WHERE m.person_id = p.person_id
-> );
```

first_name	last_name
Muhammad	Khan
Fatima	Khan
Ali	Khan
Ayesha	Khan
Hassan	Khan
Ahmed	Ali
Noor	Ali
Karim	Ali
Sara	Ali
Rashid	Hassan
Zainab	Hassan
Malik	Hassan
Hina	Hassan
Nasir	Hussain
Amina	Hussain
Imran	Hussain
Leila	Hussain
Samir	Khan
Maha	Khan
Bashir	Khan
Waqar	Ahmed
Rania	Ahmed
Khalid	Ahmed
Dina	Ahmed
Jamal	Malik
Lia	Malik
Amir	Malik
Tariq	Hassan
Nadia	Hassan

Identifies stable citizens who have zero documented history of changing districts.

```
mysql> SELECT p.first_name, p.last_name
-> FROM Person p
-> WHERE NOT EXISTS (
->     SELECT 1
->     FROM Migration_History m
->     WHERE m.person_id = p.person_id
-> );
```

first_name	last_name
Farah	Khan
Zahir	Ahmad
Meena	Ahmad
Kamil	Ahmad
Ibrahim	Raza
Saira	Raza
Kashan	Raza
Babar	Khan
Mariam	Khan
Danial	Khan
Habib	Malik
Zara	Malik
Karim	Malik
Faisal	Hassan
Iqra	Hassan
Hamza	Hassan
Salim	Ahmed
Saba	Ahmed
Hassam	Ahmed
Riaz	Ali
Bushra	Ali
Imad	Ali
Nadim	Khan
Suhana	Khan
Shiraz	Khan
Qasim	Malik
Hafsa	Malik

C. Aggregate Queries

Calculates the total headcount of recorded citizens in the census system database.

```
mysql> SELECT COUNT(*) AS TotalPopulation
-> FROM Person;
+-----+
| TotalPopulation |
+-----+
|          102   |
+-----+
1 row in set (0.01 sec)
```

Computes the gross combined monthly income of the entire surveyed population.

```
mysql> SELECT SUM(monthly_income) AS TotalIncome
-> FROM Person;
+-----+
| TotalIncome |
+-----+
| 2149000.00  |
+-----+
1 row in set (0.00 sec)
```

Evaluates the national baseline average monthly income across all citizens.

```
mysql> SELECT AVG(monthly_income) AS AverageIncome
-> FROM Person;
+-----+
| AverageIncome |
+-----+
| 30700.000000  |
+-----+
1 row in set (0.00 sec)
```

Pinpoints the absolute lowest recorded individual monthly income entry.

```
mysql>
mysql> SELECT MIN(monthly_income) AS LowestIncome
      -> FROM Person;
+-----+
| LowestIncome |
+-----+
|          0.00 |
+-----+
1 row in set (0.00 sec)
```

Pinpoints the absolute highest recorded individual monthly income entry.

```
mysql> SELECT MAX(monthly_income) AS HighestIncome
      -> FROM Person;
+-----+
| HighestIncome |
+-----+
|       72000.00 |
+-----+
1 row in set (0.00 sec)
```

Groups and counts total population distribution cleanly across distinct biological sexes.

```
mysql> SELECT b.sex_name, COUNT(*) AS Population
      -> FROM Person p
      -> INNER JOIN Biological_Sex b ON p.sex_id = b.sex_id
      -> GROUP BY b.sex_name;
+-----+-----+
| sex_name | Population |
+-----+-----+
| Male     |          64 |
| Female   |          38 |
+-----+-----+
2 rows in set (0.00 sec)
```

Filters demographic categories to show only biological sexes with more than 40 individuals.

```
mysql> SELECT sex_id, COUNT(*) AS Population
-> FROM Person
-> GROUP BY sex_id
-> HAVING COUNT(*) > 40;

+-----+-----+
| sex_id | Population |
+-----+-----+
|      1 |          64 |
+-----+-----+
1 row in set (0.00 sec)
```

Breaks down population counts and average monthly income metrics categorized by religion.

```
mysql> SELECT r.religion_name, COUNT(*) AS Population, AVG(monthly_income) AS AvgIncome
-> FROM Person p
-> INNER JOIN Religion r ON p.religion_id = r.religion_id
-> GROUP BY r.religion_name;

+-----+-----+-----+
| religion_name | Population | AvgIncome |
+-----+-----+-----+
| Islam        |          93 | 29984.375000 |
| Hinduism     |           6 | 32000.000000 |
| Christianity  |           3 | 51000.000000 |
+-----+-----+-----+
3 rows in set (0.01 sec)
```

Combines distinct first names from both citizens and field staff, removing duplicate values.

```
mysql> SELECT first_name, last_name FROM Person
-> UNION
-> SELECT first_name, last_name FROM Census_Staff;
```

first_name	last_name
Muhammad	Khan
Fatima	Khan
Ali	Khan
Ayesha	Khan
Hassan	Khan
Ahmed	Ali
Noor	Ali
Karim	Ali
Sara	Ali
Rashid	Hassan
Zainab	Hassan
Malik	Hassan
Hina	Hassan
Nasir	Hussain
Amina	Hussain
Imran	Hussain
Leila	Hussain
Samir	Khan
Maha	Khan
Bashir	Khan
Waqar	Ahmed
Rania	Ahmed
Khalid	Ahmed
Dina	Ahmed
Jamal	Malik
Lia	Malik
Amir	Malik
Tariq	Hassan
Nadia	Hassan
Farhan	Hassan

Combines all first names from citizens and field staff, preserving every duplicate entry.

```
mysql> SELECT first_name, last_name FROM Person
-> UNION ALL
-> SELECT first_name, last_name FROM Census_Staff;
```

first_name	last_name
Muhammad	Khan
Fatima	Khan
Ali	Khan
Ayesha	Khan
Hassan	Khan
Ahmed	Ali
Noor	Ali
Karim	Ali
Sara	Ali
Rashid	Hassan
Zainab	Hassan
Malik	Hassan
Hina	Hassan
Nasir	Hussain
Amina	Hussain
Imran	Hussain
Leila	Hussain
Samir	Khan
Maha	Khan
Bashir	Khan
Waqar	Ahmed
Rania	Ahmed
Khalid	Ahmed
Dina	Ahmed
Jamal	Malik
Lia	Malik
Amir	Malik
Tariq	Hassan
Nadia	Hassan
Farhan	Hassan
Yousef	Ali
Hana	Ali

Finds common names shared by both active citizens and field census staff.

```
mysql> SELECT first_name, last_name
-> FROM Person
-> WHERE first_name IN (
->     SELECT first_name
->     FROM Census_Staff
-> );
```

first_name	last_name
Fatima	Khan
Ali	Khan
Ahmed	Ali
Zainab	Hassan
Hina	Hassan
Imran	Hussain
Tariq	Hassan
Nadia	Hassan
Adnan	Ahmed
Hamza	Hassan
Saba	Ahmed
Bushra	Ali
Rabia	Hassan
Rizwan	Ali

14 rows in set (0.01 sec)

Lists first names unique to citizens that do not exist among census staff.

```
mysql> SELECT first_name, last_name
-> FROM Person
-> WHERE first_name NOT IN (
->     SELECT first_name
->     FROM Census_Staff
-> );
```

first_name	last_name
Muhammad	Khan
Ayesha	Khan
Hassan	Khan
Noor	Ali
Karim	Ali
Sara	Ali
Rashid	Hassan
Malik	Hassan
Nasir	Hussain
Amina	Hussain
Leila	Hussain
Samir	Khan
Maha	Khan
Bashir	Khan
Waqar	Ahmed
Rania	Ahmed
Khalid	Ahmed
Dina	Ahmed
Jamal	Malik
Lia	Malik
Amir	Malik
Farhan	Hassan
Yousef	Ali
Hana	Ali
Rasheed	Ali
Saad	Khan
Luna	Khan
Aziz	Khan
Yasmin	Ahmed

D. Views

Shows a complete profile of each citizen, including their full name, age, education, job, and income in plain words instead of codes.

```
mysql> SELECT * FROM Citizen_SocioEconomic_Profile;
```

person_id	full_name	cnic_number	biological_sex	age	religion	education_level	employment_status	occupation	industry	monthly_income
1	Muhammad Khan	12345-6000001-1	Male	61	Talim	Masters	Employed Full-Time	Doctor/Medical Professional	Public Administration	45000.00
2	Fatima Khan	12345-6000002-3	Female	56	Islam	PHD	Homemaker	NULL	NULL	0.00
3	Ali Khan	12345-6000003-5	Male	30	Talim	Masters	Student	Government Official	Wholesale and Retail Trade	0.00
4	Ayesha Khan	12345-6000004-7	Female	27	Islam	Intermediate	Student	NULL	NULL	0.00
5	Hassan Khan	12345-6000005-9	Male	24	Talim	Primary	Student	NULL	NULL	0.00
6	Ahmed Ali	12345-6000006-0	Male	57	Islam	Matriculation	Employed Full-Time	Driver/Transport Worker	Agriculture and Farming	38000.00
7	Noor Ali	12345-6000007-2	Female	53	Talim	Middle	Homemaker	NULL	NULL	0.00
8	Karim Ali	12345-6000008-4	Male	32	Islam	Intermediate	Employed Full-Time	Shopkeeper/Retail Merchant	Health and Social Work	32000.00
9	Sara Ali	12345-6000009-6	Female	25	Talim	Primary	Student	NULL	NULL	0.00
10	Rashid Hassan	12345-6000010-8	Male	66	Islam	Middle	Employed Full-Time	Crop Farm Laborer	Agriculture and Farming	42000.00
11	Zainab Hassan	12345-6000011-0	Female	60	Talim	Primary	Homemaker	NULL	NULL	0.00
12	Malik Hassan	12345-6000012-2	Male	33	Islam	PHD	Employed Full-Time	Primary School Teacher	Utilities (Water/Electricity)	35000.00
13	Hina Hassan	12345-6000013-4	Female	26	Talim	Matriculation	Student	NULL	NULL	0.00
14	Nasir Hussain	12345-6000014-6	Male	63	Islam	Middle	Employed Full-Time	Construction Laborer	Construction	48000.00
15	Aamina Hussain	12345-6000015-8	Female	59	Talim	Uneducated	Homemaker	NULL	NULL	0.00
16	Imran Hussain	12345-6000016-1	Male	30	Islam	Matriculation	Employed Full-Time	Domestic Helper/Maid	Public Administration	28000.00
17	Leila Hussain	12345-6000017-3	Female	22	Talim	Primary	Student	NULL	NULL	0.00
18	Samir Khan	12345-6000018-5	Male	56	Islam	Matriculation	Employed Full-Time	Factory Worker	Manufacturing - Textiles	52000.00
19	Maha Khan	12345-6000019-7	Female	50	Talim	Middle	Homemaker	NULL	NULL	0.00
20	Bashir Khan	12345-6000020-9	Male	28	Islam	Intermediate	Student	Engineer	Wholesale and Retail Trade	0.00
21	Wagdy Ahmed	12345-6000021-0	Male	70	Talim	Primary	Unemployed - Not Seeking Work	Software Developer	Financial Services	65000.00
22	Rania Ahmed	12345-6000022-2	Female	66	Islam	Primary	Homemaker	NULL	NULL	0.00
23	Khalid Ahmed	12345-6000023-4	Male	40	Talim	Bachelors	Employed Full-Time	Tailor/Seamstress	Transportation and Logistics	55000.00
24	Dina Ahmed	12345-6000024-6	Female	38	Islam	Intermediate	Employed Full-Time	NULL	NULL	25000.00
25	Jamal Malik	12345-6000025-8	Male	32	Islam	Matriculation	Employed Full-Time	Business Owner	Telecommunications	41000.00
26	Lia Malik	12345-6000026-1	Female	47	Islam	Uneducated	Homemaker	NULL	NULL	0.00
27	Amir Malik	12345-6000027-3	Male	25	Talim	Middle	Student	NULL	NULL	0.00
28	Tariq Hassan	12345-6000028-5	Male	62	Islam	Middle	Employed Full-Time	Agricultural Farmer	Hospitality and Tourism	39000.00
29	Nadia Hassan	12345-6000029-7	Female	57	Talim	Primary	Homemaker	NULL	NULL	0.00
30	Farhan Hassan	12345-6000030-9	Male	32	Islam	Intermediate	Employed Full-Time	Driver/Transport Worker	Education	31000.00
31	Yusef Ali	12345-6000031-0	Male	68	Talim	Primary	Unemployed - Not Seeking Work	Shopkeeper/Retail Merchant	Agriculture and Farming	72000.00
32	Hana Ali	12345-6000032-2	Female	62	Islam	Primary	Homemaker	NULL	NULL	0.00
33	Rasheed Ali	12345-6000033-4	Male	35	Talim	Matriculation	Employed Full-Time	Nurse	Wholesale and Retail Trade	29000.00
34	Saad Khan	12345-6000034-6	Male	60	Islam	Middle	Employed Full-Time	Doctor/Medical Professional	Public Administration	37000.00
35	Luna Khan	12345-6000035-8	Female	54	Talim	Primary	Homemaker	NULL	NULL	0.00
36	Aziz Khan	12345-6000036-1	Male	27	Islam	Middle	Student	NULL	NULL	0.00
37	Adnan Ahmed	12345-6000037-3	Male	64	Talim	Middle	Employed Full-Time	Construction Laborer	Utilities (Water/Electricity)	46000.00
38	Yasmin Ahmed	12345-6000038-5	Female	60	Islam	Uneducated	Homemaker	NULL	NULL	0.00
39	Rana Ahmed	12345-6000039-7	Male	29	Talim	Matriculation	Employed Full-Time	Domestic Helper/Maid	Health and Social Work	24000.00
40	Wasim Malik	12345-6000040-9	Male	56	Islam	Matriculation	Employed Full-Time	Agricultural Farmer	Manufacturing - Textiles	51000.00
41	Rana Malik	12345-6000041-0	Female	51	Talim	Middle	Homemaker	NULL	NULL	0.00
42	Naeem Malik	12345-6000042-2	Male	25	Islam	Primary	Student	NULL	NULL	0.00
43	Mohan Hassan	12345-6000043-4	Male	69	Hinduism	Primary	Employed Full-Time	Software Developer	Agriculture and Farming	58000.00
44	Anjali Hassan	12345-6000044-6	Female	63	Hinduism	Uneducated	Homemaker	NULL	NULL	0.00
45	Rohan Hassan	12345-6000045-8	Male	34	Hinduism	Intermediate	Employed Full-Time	Electrician	Public Administration	26000.00
46	Deepak Ali	12345-6000046-1	Male	62	Hinduism	Middle	Employed Full-Time	Business Owner	Hospitality and Tourism	44000.00
47	Priya Ali	12345-6000047-3	Female	58	Hinduism	Primary	Homemaker	NULL	NULL	0.00
48	Azmun Ali	12345-6000048-5	Male	29	Hinduism	Matriculation	Student	NULL	NULL	0.00
49	Siddiqui Khan	12345-6000049-7	Male	64	Talim	Middle	Employed Full-Time	Crop Farm Laborer	Agriculture and Farming	35000.00
50	Gulnar Khan	12345-6000050-9	Female	59	Islam	Uneducated	Homemaker	NULL	NULL	0.00
51	Farah Khan	12345-6000051-0	Female	33	Talim	Matriculation	Employed Full-Time	NULL	NULL	18000.00
52	Zahir Ahmad	12345-6000052-2	Male	65	Islam	Primary	Employed Full-Time	Driver/Transport Worker	Wholesale and Retail Trade	62000.00
53	Meena Ahmad	12345-6000053-4	Female	60	Talim	Primary	Homemaker	NULL	NULL	0.00
54	Kamil Ahmad	12345-6000054-6	Male	30	Islam	Intermediate	Employed Full-Time	Plumber	Manufacturing - Other	33000.00
55	Ibrahim Raza	12345-6000055-8	Male	66	Talim	Middle	Employed Full-Time	Doctor/Medical Professional	Construction	40000.00
56	Saira Raza	12345-6000056-1	Female	62	Islam	Primary	Homemaker	NULL	NULL	0.00
57	Rachna Raza	12345-6000057-3	Male	31	Talim	Matriculation	Employed Full-Time	Factory Worker	Manufacturing - Textiles	23000.00

Shows the living conditions of each home, tracking things like room counts, clean water access, electricity, and whether it is in a city or a village.

```
mysql> SELECT * FROM HouseholdLivingStandards;
mysql> SELECT * FROM HouseholdLivingStandards
  -- WHERE area_classification = 'Rural';
```

household_size	number_of_rooms	census_block_code	area_classification	structure_type	water_supply	lighting_source	cooking_fuel	washroom_type	kitchen_type
4	2	R1P-07004	Rural	Pure Residential	Motor/Borehole	Electricity Grid	Natural Gas Piping	Flush connected to septic tank	Separate indoor kitchen
4	3	R1P-07006	Rural	Pure Residential	Tap Water	Electricity Grid	Natural Gas Piping	Flush connected to public sewerage	Separate indoor kitchen
3	4	R1P-07008	Rural	Mixed Use Residential Commercial	Hand Pump	Solar Power	Firewood	Fit Latrine	Outdoor cooking space
3	3	R1P-07010	Rural	Mixed Use Residential Commercial	Tap Water	Electricity Grid	Natural Gas Piping	Flush connected to public sewerage	Separate indoor kitchen
3	2	R1P-07004	Rural	Pure Residential	Hand Pump	Solar Power	Firewood	Fit Latrine	Outdoor cooking space
3	2	R1P-07016	Rural	Pure Residential	Motor/Borehole	Electricity Grid	LPG Cylinders	Flush connected to septic tank	Separate indoor kitchen
3	3	R1P-07018	Rural	Commercial Only	Tap Water	Electricity Grid	Natural Gas Piping	Flush connected to public sewerage	No kitchen space

7 rows in set (0.00 sec)

Groups Pakistan's geographic areas together to show the total number of census blocks and households in every region at a glance.

```
mysql> SELECT province_name, district_name, SUM(gross_estimated_households) AS total_households FROM Regional_Administrative_Summary GROUP BY province_name, district_name ORDER BY total_households DESC;
```

province_name	district_name	total_households
Sindh	Karachi Central	3440
Sindh	Karachi South	3350
Islamabad Capital Territory	Islamabad Urban	2800
Sindh	Karachi East	2650
Sindh	Hyderabad	1910
Balochistan	Quetta	1810
Khyber Pakhtunkhwa	Baizpur	1720
Khyber Pakhtunkhwa	Nowshera	1710
Khyber Pakhtunkhwa	Mardan	1660
Khyber Pakhtunkhwa	Swabi	1600
Khyber Pakhtunkhwa	Peshawar	1520
Khyber Pakhtunkhwa	Swat	1500
Khyber Pakhtunkhwa	Abbottabad	1450
Sindh	Larkana	1450
Sindh	Shikarpur	1440
Sindh	Sukkur	1410
Punjab	Lahore	1400
Punjab	Faisalabad	1390
Sindh	Jamshoro	1370
Gilgit Baltistan	Gilgit	1350
Punjab	Aitook	1320
Punjab	Zhang	1255
Punjab	Multan	1230
Punjab	Bahawalpur	1215
Punjab	Rawalpindi	1210
Balochistan	Kech (Turbat)	1190
Khyber Pakhtunkhwa	Mansehra	1160
Punjab	Sheikhpura	1150
Sindh	Jacobabad	1090
Balochistan	Khuzdar	1080
Khyber Pakhtunkhwa	Charsadda	1060
Sindh	Khoitpur	1060
Islamabad Capital Territory	Islamabad Rural	1050
Khyber Pakhtunkhwa	Chitral	1040
Balochistan	Kalat	1020
Sindh	Dadu	1000
Sindh	Ghotki	990
Balochistan	Panipur	980
Balochistan	Qwadar	950
Punjab	Rahim Yar Khan	945
Punjab	Vehari	935
Punjab	Bahawalnagar	925
Punjab	Khanewal	910
Islamabad Capital Territory	Capital Territory East	900
Punjab	Toba Tek Singh	875
Khyber Pakhtunkhwa	Lower Dir	870
Balochistan	Pishin	870
Balochistan	Mastung	840
Punjab	Kasur	810
Punjab	Jhelum	795
Balochistan	Chaman	780
Gilgit Baltistan	Banara	710
Gilgit Baltistan	Nagar	660
Khyber Pakhtunkhwa	Rustan (Proposed)	660

54 rows in set (0.31 sec)

10. Cascading Operations

• ON DELETE CASCADE

In our census architecture, the **Migration_History** table acts as a tracking log for changing districts. It is completely dependent on a person's profile existing in the **Person** table. If a citizen is legally removed from the census database (due to data entry errors or fraud cancellation), keeping their old migration history is meaningless and leaves orphaned logs.

By applying ON DELETE CASCADE, removing a record from the Person table automatically triggers a chain reaction that deletes all corresponding relocation entries from Migration_History.

• ON DELETE SET NULL

Our census operations utilize a supervisor network in the **Census_Staff** table. Field enumerators are monitored via a recursive self-referencing foreign key column called supervisor_id. If a supervisor resigns, retires, or is removed from the staff workforce database, we absolutely **do not** want to lose or delete the subordinate field enumerators they managed.

By using ON DELETE SET NULL, when a supervisor's record is removed, the field enumerators they were managing remain perfectly intact in the database; their supervisor_id simply switches to NULL (meaning they are temporarily unassigned until a new manager takes over).

11. Challenges and Risks

1. Handling Large Data Efficiently

- **The Challenge:** A national census involves capturing records for tens of millions of citizens and millions of households. Standard un-normalized or flat-file database architectures experience catastrophic performance degradation and massive disk space bloating when handling data at this scale.
- **Our Strategy (Normalization & Storage Optimization):**
 - **Elimination of Redundant Text:** Instead of storing repetitive textual data (e.g., storing the string "Tap Water" or "Employed Full-Time" millions of times across individual household and person rows), these attributes were extracted into micro-lookup tables (Water_Source, Employment_Status, etc.).
 - **Strategic Use of Compact Data Types:** By linking parent tracking records via **TINYINT** (1 byte per row, supporting up to 255 distinct lookup categories) instead of INT (4 bytes) or VARCHAR text strings, our schema dramatically compresses the overall database footprint. When scaled to 200 million citizen records, saving 3 bytes per column across dozens of columns prevents gigabytes of unnecessary data footprint, ensuring data blocks fit efficiently into system cache memory during intense analytical processing.
 - **Achieving 3rd Normal Form (3NF):** The database schema was systematically advanced to **3NF**. Every non-prime attribute depends purely, directly, and non-transitively on the primary key anchor of its respective table, ensuring computational efficiency and keeping data storage structurally lean.

2. Ensuring Referential Integrity

- **The Challenge:** In a highly relational database with deep organizational hierarchies (e.g., Province --> Division --> District --> Tehsil --> Union_Council --> Census_Block --> Structure --> Household --> Person), deleting or updating a higher-level record risks creating orphaned child data blocks and breaking internal consistency.
- **Our Strategy:**
 - Implemented structural relational rules via declarative foreign keys.
 - Applied **ON DELETE CASCADE** on strict target operational tracking tables (such as wiping out a citizen's internal logs in Migration_History automatically if their baseline Person profile is deleted).
 - Applied **ON DELETE SET NULL** on organizational management linkages (such as self-referencing workforce hierarchies in Census_Staff), ensuring field enumerator operational performance data is preserved if a supervisor's account is removed.

3. Handling Complex Queries and Optimization

- **The Challenge:** As the volume of deep relational tables grows into millions of data points, executing multi-table statements utilizing numerous INNER JOIN, LEFT JOIN, or nested aggregate operations can lead to slow performance due to intensive nested loop evaluations or extensive full table scans.
- **Our Strategy:**
 - **Relational Views:** Wrapped multi-tier joins inside predefined logical relational SQL views (Most_Populated_Cities, Highly_Educated_Persons, Citizen_Record). This abstracts structural execution complexities and helps the database engine optimize compilation paths.

Group Members & Project Roles

Members	Roles
Azma	Wrote sub queries, co-related queries and aggregate functions
Sarmad	Developed multi-table Joins and added Views
Sharjeel	Designed the schema, data insertion and documentation
Zakia	Wrote sub queries, co-related queries and aggregate functions